

Hieronym: Leveraging Hierarchical Multi-Source Information for Function Renaming in Stripped Binary

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Abstract

Function renaming in stripped binaries can substantially assist reverse engineers by improving code readability, yet it is a challenging task. The difficulty stems from the need to accurately capture function semantics from low-level binary code across diverse instruction sets, architectures, and compiler optimizations, and to express these semantics in concise, human-readable names. Existing approaches either inadequately capture comprehensive function semantics or exhibit limited generalization to previously unseen binaries. In this paper, we present HIERONYM, a generative large language model (LLM)-based framework for stripped binary function renaming. HIERONYM adopts a hierarchical summarization-driven domain adaptation strategy and integrates multi-source information—including global binary context, local calling context, and intrinsic function semantics—to enhance the LLM’s understanding of binary code. To enable systematic evaluation, we further propose a dual-layer evaluation framework that incorporates both token-level and whole-name-level metrics. We evaluate HIERONYM on binary functions compiled with four compiler optimization levels (O0–O3) for four architectures (x64, x86, ARM, and MIPS). Experimental results demonstrate that HIERONYM significantly outperforms state-of-the-art methods, achieving token-level improvements of 50.12% in precision, 41.75% in recall, and 45.10% in F1-score, as well as a 79.94% improvement in name-level accuracy, while also exhibiting strong generalization capability. Moreover, experiments on real-world malware samples further validate the practical effectiveness of HIERONYM in security-critical scenarios.

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• Computing methodologies → Machine learning; • Security and privacy → Software reverse engineering.

Keywords

Reverse Engineering; Large Language Models; Binary Function Renaming; Program Comprehension

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1 Introduction

Reverse engineering is a fundamental technique in security analysis, supporting tasks such as vulnerability discovery [59, 65], malware analysis [2, 67], and security auditing [21]. In practice, analysts frequently encounter stripped binaries lacking high-level information, including function and variable names, structure details and optimization information. This loss obscures program intent and hinders analysis efficiency. Among the missing information, function names are particularly critical, as they provide concise abstractions of program functionality and serve as key cognitive anchors during program comprehension [9, 57]. However, widely used decompilation tools such as IDA Pro and Ghidra typically assign address-based placeholder names (such as FUN_00129f48) that convey little information. Accurate function naming still depends heavily on manual expert analysis [68], rendering the process time-consuming and labor-intensive. Consequently, automatic function renaming for stripped binaries has emerged as an important research direction.

However, automatic function renaming is a challenging task, as it requires both a comprehensive understanding of binary function semantics and effective abstraction of such semantics into concise and meaningful natural-language keywords. Recent machine learning (ML)-based approaches have shown promise [18, 25, 36, 45, 46, 69]. These methods typically employ weak base models, such as LSTM [73] in NFRE [18] and BERT [13] in SymLM [36], to understand function semantics and adopt a classification-based paradigm that maps function semantics to a predefined, closed vocabulary. These models often struggle to capture function semantics robustly, and exhibit limited ability to abstract function semantics into natural-language keywords. While they may perform well on training data, they often lack sufficient generalization to unseen binaries. Generative large language models (LLMs) have recently demonstrated strong generalization and reasoning capabilities, achieving impressive results in code understanding and generation. These advances have spurred growing interest in applying LLMs to binary function renaming. However, despite achieving state-of-the-art (SOTA) performance, existing LLM-based methods still fall short of practical expectations: even the best-performing approach achieves only a 0.277 F1-score. This discrepancy raises a critical question: **why have existing methods not delivered the performance we would expect from LLMs in this setting?** We analyze this gap and identify the following challenges.

C1: Flawed Evaluation Metrics. Prior evaluation metrics for function renaming are flawed, as they sometimes require models to reproduce fine-grained details unavailable from binary function alone. For example, the prefix in `jitterc_get_lineno` encodes project-specific context unrecoverable from the binary functions alone. Penalizing the prediction of `get_current_line_number` is therefore unjustified. Moreover, most evaluations rely on token-level metrics to mitigate the out-of-vocabulary (OOV) issue, but this choice is fragile. Erroneous tokenization can drastically distort evaluation metrics. For instance, SymGen [33] tokenizes `c_isascii_string` into `{c, isa, sc, ii, string}`, reducing its F1-score from 0.857 to 0.25. Since a sound evaluation is essential for revealing the true limitations of existing methods and supporting a faithful interpretation of their performance, how to properly evaluate the performance of the function renaming task is a non-trivial challenge.

C2: Semantic Sparsity of Stripped Binaries. Motivated by C1, we observe that stripped binary functions contain only limited semantic information, which often makes accurate name recovery unattainable. For example, the `jitterc`-specific identifier discussed above is essentially impossible to recover from a single stripped binary function. Although prior works, such as SymLM [36], XFL [46], and BLens [4], attempt to incorporate additional context, such as calling context, this information is often still insufficient to compensate for the loss of semantics in stripped binaries, especially for context-specific identifiers. Moreover, the contextual representations used by these methods are sensitive to architectural and optimization variations, which limits their robustness. Therefore, a key challenge is how to model essential, task-relevant context from information-limited binaries while remaining robust to architecture- and optimization-induced changes.

C3: Knowledge Gaps Between Natural Language and Binary Code Semantics in LLMs. Although LLMs are strong at understanding and generating source code, their training corpora are

dominated by natural language and include relatively little code, which leaves them with limited ability to understand stripped binary [47, 55]. This mismatch creates a substantial gap between the knowledge encoded in general-purpose LLMs and the understanding required for binary analysis. A straightforward approach to mitigating this gap is domain adaptation. Accordingly, function summarization-based domain adaptation strategies have been explored and shown some effectiveness in transferring LLMs to the binary domain [33]. However, prior studies indicate that function-level summaries alone are insufficient to capture deep and comprehensive function semantics [22, 29–31, 44]. Consequently, effectively bridging the knowledge gap between general-purpose LLMs and the binary domain remains an urgent open problem.

In this paper, we present HIERONYM, a novel generative framework for function renaming in stripped binaries. It is designed to learn hierarchical binary function semantics and to comprehensively model binary functions by integrating multi-source information. Leveraging generative LLMs, HIERONYM abstracts binary semantics into natural-language keywords. To ensure comprehensive and reliable evaluation (C1), we adopt a dual-layer evaluation framework that assesses performance at both the token and name levels. We propose an enhanced token-level evaluation with a hybrid tokenization mechanism to produce semantically meaningful tokens to mitigate tokenization errors, along with heuristic rules to measure token-level semantic relatedness. Our evaluation framework is further completed by a name-level evaluation using an LLM-as-a-judge approach to assess whole-name semantic equivalence. To comprehensively model function semantics (C2), HIERONYM integrates three complementary information sources: global binary context, local calling context, and the target function itself. Global and local contexts provide essential task-relevant information, which we represent using textual summaries. Our representation offers a robust abstraction that is resilient to code variations introduced by different compilation. To bridge the knowledge gap between general-purpose LLMs and the binary domain (C3), we introduce hierarchical function summarization as a domain adaptation strategy, enabling multi-view semantic understanding of functions.

Our comprehensive evaluation demonstrates that HIERONYM exhibits exceptional generalizability and superior performance compared to SOTA function renaming methods, including SymLM [36], XFL [46], BLens [4], and SymGen [33], across four instruction set architectures (x64, x86, ARM, and MIPS) and four optimization levels (O0–O3). HIERONYM achieves substantial gains in token-level precision, recall, and F1-score of 50.12%, 41.75%, and 45.10%, respectively, as well as a 79.94% improvement in name-level accuracy over the best prior method. Leveraging the multi-source information in our approach, HIERONYM even successfully recovers the `jitterc`-specific identifier discussed above. Ablation studies further validate the effectiveness of HIERONYM's design. Hierarchical function summarization based domain adaptation improves F1-score by 18.83%, while incorporating global binary context and local calling context yields additional gains of 27.63% and 13.81%, respectively. These results highlight the contributions of each component and underscore the novelty and effectiveness of the proposed framework. Finally, experiments on real-world malware binaries confirm HIERONYM's practical utility for security analysis.

The contributions of this paper can be summarized as follows:

- We propose a dual-layer evaluation framework that evaluates performance at both the token-level and the name-level, enabling a reliable evaluation of function renaming methods.
- We propose HIERONYM, a novel generative approach for function renaming in stripped binaries that leverages multi-source information. By integrating global binary context, local calling context, and target function, HIERONYM enables LLMs to learn comprehensive function semantics, resulting in improved performance and generalizability.
- We introduce a hierarchical function summarization-based domain adaptation strategy that effectively adapts pre-trained generative LLMs to binary semantic modeling.
- Extensive experiments show that HIERONYM consistently outperforms SOTA methods in function renaming. The results further highlight its strong generalizability, the effectiveness of individual components, and its practical applicability. We make our code and dataset publicly available at: <https://github.com/NASP-THU/Hieronym>.

2 Background and Motivation

2.1 Problem Definition

Function renaming in stripped binaries requires both accurate reverse engineering of binary semantics and the generation of names that faithfully reflect those semantics. Decoder-only LLMs, based on the Transformer decoder architecture [47, 55], have demonstrated strong capabilities in text understanding and generation, particularly exhibiting impressive generalization in zero- and few-shot settings [6]. In contrast to existing ML-based approaches [18, 36, 69], which formulate function renaming as a classification task over a closed vocabulary, we follow SymGen [33] and model function renaming as an autoregressive generation task using LLMs. Formally, given a binary function $\mathcal{F}$, represented as a sequence of binary code tokens $\mathcal{F} = \{t_1, t_2, \dots, t_n\}$, a function renaming model $\mathcal{M}$ generates a sequence of natural-language tokens $\mathcal{W} = \{w_1, w_2, \dots, w_m\}$, i.e., $\mathcal{M} : \mathcal{F} \rightarrow \mathcal{W}$. Under the autoregressive paradigm, the generation probability of $\mathcal{W}$ is factorized as

$$p(\mathcal{W}) = \prod_{j=1}^m p(w_j \mid w_1, \dots, w_{j-1}, \mathcal{F}), \quad (1)$$

where each token w_j is conditioned on all previously generated tokens and the binary function semantics.

2.2 Existing Techniques and Our Motivations

Although many studies have explored function renaming in stripped binaries, prior efforts have not adequately addressed the challenges outlined above, which directly motivates our proposal of HIERONYM.

Existing methods primarily evaluate function-renaming performance at the token level, yet the reliability of such evaluation heavily depends on the underlying tokenization strategy, which directly affects subsequent semantic alignment. Many methods—including XFL [46], BLens [4], SymLM [36], and SymGen [33]—rely on rule-based tokenization, which often introduces segmentation bias and degrades evaluation metrics. For example, SymLM incorrectly splits `test_nofork_sideeffects` into {t, est, no, for, k, side, eff, ect,

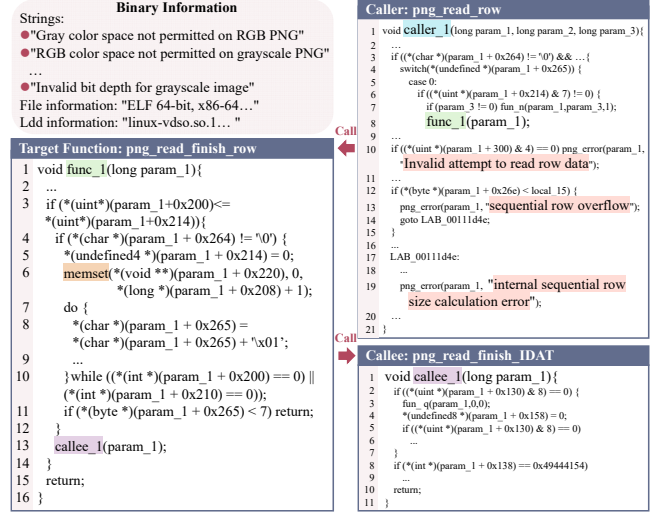


Figure 1: An Example Illustrating the Importance of Task-Related Information in Function Naming

s}, hindering correct alignment with the ground-truth tokens. Semantic alignment based on tokenization results also remains limited. XFL and BLens depend on general English synonym sets that fail to capture domain-specific morphological variations, while the CodeWordNet [36] models used by SymLM and SymGen are trained on small corpora with insufficient coverage of OOV tokens. As a result, existing token-level evaluation strategies suffer from unreliable tokenization and weak semantic alignment, and the absence of name-level evaluation further prevents current methods from systematically addressing C1.

Grasping task-relevant contextual information is essential for modeling function semantics and improving function-renaming performance. Figure 1 illustrates this through a real-world example of a stripped binary, showing the decompiled code of a target function together with its callers, callees, and domain-related information extracted from binary (simplified for clarity). The example involves three functions—`func_1`, `caller_1`, and `callee_1`—whose ground-truth names are `png_read_finish_row`, `png_read_row`, and `png_read_finish_IDAT`, respectively. Although decompiled code exposes richer semantics than raw binaries, the target function alone is insufficient for accurate name inference. As shown in Figure 1, the semantics of `png_read_finish_row` are tightly coupled with both its caller and callee. Prior works have attempted to incorporate calling context in different ways. SymLM models calling context using assembly-level representations, while XFL and BLens rely on handcrafted features to capture calling relationships. However, these methods are sensitive to architecture and optimization settings, causing unstable semantics (e.g., SymLM reports a 60.2% F1-score drop from O3 to O0). Domain-related information (e.g., the keyword “PNG”) provides essential task-relevant cues unavailable from the local function alone. Despite its importance, existing methods do not consider such domain-related information. The LLM-based method SymGen ignores both domain information and

calling context. Existing methods lack effective mechanisms for leveraging task-relevant information and struggle to address C2.

Many existing methods, such as SymLM, BLens, and XFL, are built on traditional ML frameworks and lack intrinsic mechanisms to address C3. SymGen introduces function-level summarization to equip LLMs with binary code knowledge. However, prior studies show that function-level summaries alone are insufficient to capture fine-grained implementation details. In particular, statement-level summarization has been shown to effectively complement function-level semantics by modeling the distinct logical operations within a function [22, 31]. These two levels of abstraction are inherently complementary: statement-level summaries capture concrete implementation logic, while function-level summaries convey overall functionality. Together, they provide a more complete and nuanced understanding of code structure and semantics [22, 29]. As a result, SymGen remains limited in modeling fine-grained binary semantics and does not fully address C3.

3 Evaluation Framework

To comprehensively evaluate function renaming methods, we propose a dual-level evaluation framework: (1) **token-level evaluation**, which is widely used in prior work to mitigate OOV issue; and (2) **name-level evaluation**, which measures the correctness of generated function names from a holistic semantic perspective. By combining these two complementary levels, our framework enables a systematic evaluation of both fine-grained semantic recognition and overall semantic expressiveness.

3.1 Token-Level Evaluation

Token-level evaluation typically consists of two stages: (1) function-name tokenization, which segments a name into meaningful tokens, and (2) token semantic alignment, which determines semantic relationships among tokens. In the tokenization stage, we adopt a hybrid strategy that combines rule-based heuristics with LLM assistance. In the subsequent alignment stage, heuristic rules are applied to identify semantic associations between tokens.

Name Tokenization. Function names are typically multi-word natural-language expressions that include abbreviations, domain-specific jargon, and language-dependent conventions [27, 49]. Unlike natural language text with explicit word boundaries, code supports diverse naming styles. Although capitalization, underscores, or digits are commonly used as delimiters, they are often omitted, resulting in unsegmented character sequences with ambiguous word boundaries. For function names that follow conventional naming styles—such as camelCase, underscore-separated, or numeric-delimited forms—we apply a rule-based tokenization strategy. For unconventional names, rule-based methods used in prior work [33, 36] tend to over-segment names, breaking coherent semantic tokens and thereby undermining token-level evaluation. To address this limitation, we propose an LLM-based few-shot tokenization approach that leverages contextual understanding to accurately segment complex function names. We provide the LLM with a small set of diverse in-context examples covering different naming styles and domain-related vocabulary to guide structured

parsing. For each function name, we generate three independent tokenization results and aggregate them using a voting-based strategy. The final segmentation is determined as follows:

- A segmentation boundary is retained if it is suggested in at least two of the three results.
- Among the three candidates, the result with the largest intersection with the retained boundary set is selected, with any redundant or overlapping labels removed.

By integrating rule-based heuristics with LLM-driven tokenization, our framework achieves robust and effective segmentation of function names, providing a reliable foundation for subsequent token-level semantic analysis.

Token Semantic Alignment. Function names exhibit substantial lexical variability due to developer preferences, including morphological variants, abbreviations, and misspellings [27, 34, 36, 49]. Ignoring such relationships significantly biases evaluation, and manual curation does not scale to large vocabularies. To address this issue, we build on prior work [18, 69] and propose an automated method for identifying semantically related tokens. We categorize token relationships into four types and handle each as follows:

- For morphological variants, we apply lemmatization to determine whether two tokens share the same lexical root.
- For abbreviation, we use Smith-Waterman algorithm [51] to compute token similarity. If the relative similarity exceeds 0.65 and both tokens are longer than two characters, they are considered to form an abbreviation-full-form relationship.
- For misspellings, we propose a Damerau-Levenshtein-based measure [3] to quantify token similarity, pairs exceeding a threshold of 0.75 are classified as spelling variants.
- For synonyms, we use Qwen3-Embedding [71] to mitigate OOV issues, trained on large-scale corpora, pairs with similarity scores above 0.8 are treated as synonyms.

By incorporating multi-dimensional association mechanism, our evaluation framework can identify semantically equivalent tokens, improving evaluation robustness in the presence of lexical diversity. The thresholds used in this mechanism are determined through empirical analysis and validated via experimental evaluation.

Token-Level Metrics. To evaluate our model and baselines, we adopt the standard token-level precision, recall, and F1-score metrics, following common practice in prior studies [4, 33, 36, 69]. These metrics quantify the overlap between tokens in the predicted and ground-truth function names. We define *true positives (TP)* as the number of correctly predicted tokens, including both exact matches and those considered equivalent under semantic alignment. *False positives (FP)* are the number of predicted tokens that do not appear in ground-truth tokens, while *false negatives (FN)* are the number of ground-truth tokens that are not predicted. Given a predicted name $\hat{W} = \{\hat{w}_1, \dots, \hat{w}_j\}$ and ground truth $W = \{w_1, \dots, w_m\}$, let $\alpha(\cdot)$ be an indicator function that returns 1 if its condition holds and 0 otherwise. We compute:

$$TP = \sum_{t=1}^j \alpha(\hat{w}_t \in W) \quad FP = \sum_{t=1}^j \alpha(\hat{w}_t \notin W) \quad FN = \sum_{t=1}^m \alpha(w_t \notin \hat{W})$$

Precision (P), recall (R), and F1-score (F1) are then computed as:

$$P = \frac{TP}{TP + FP} \quad R = \frac{TP}{TP + FN} \quad F1 = \frac{2 \times P \times R}{P + R}$$

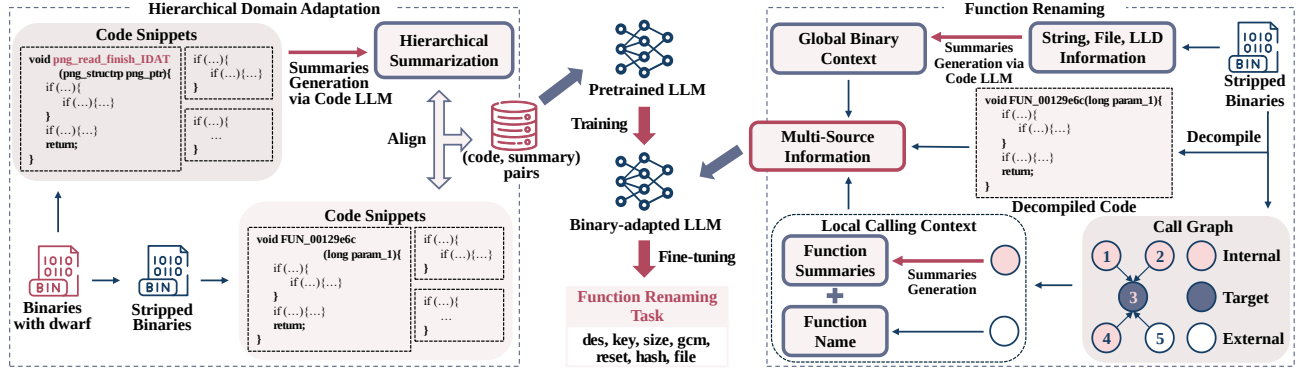


Figure 2: The Overall Workflow of HIERONYM

3.2 Name-Level Evaluation

Relying solely on token-level evaluation is insufficient for evaluating semantic consistency between predicted and ground-truth function names. Prior studies show that even for identical program semantics, developers choose the same identifier with a probability of only 6.9% [16]. Function names may be lexically dissimilar yet semantically equivalent, causing token-level metrics (e.g., F1-score) to underestimate model performance. We introduce a name-level evaluation that assesses semantic consistency from a holistic perspective to address this limitation. Recent studies indicate that LLMs used as evaluators align more closely with human judgment [52, 66]. Inspired by this finding, we adopt an LLM-assisted evaluation strategy and employ GPT-4o as a semantic judge. For each binary function, we query GPT-4o with the global binary information, its decompiled code, and a pair consisting of the ground-truth name and the predicted name. GPT-4o is instructed to determine whether the predicted name is semantically consistent with the ground-truth name and to output a binary decision (“yes” or “no”).

To mitigate potential hallucinations or subjectivity in LLM-based evaluation, we further incorporate an embedding-based semantic similarity check. We compute the cosine similarity between the predicted and ground-truth names using the Qwen3-Embedding model and apply a threshold of 0.7 determined through empirical validation. A prediction is considered semantically correct only if both the GPT-4o judgment and the embedding-based similarity criterion indicate agreement.

Name-Level Metric. We evaluate name-level performance using accuracy, defined as the proportion of predicted function names judged correct under the above criteria. Due to the scale of the dataset and the cost of LLM-based evaluation, we follow common practice [35, 66] and randomly sample 200 test functions for each architecture-optimization pair, yielding a total of 3,200 functions for name-level evaluation. This sampling strategy balances statistical reliability with computational feasibility.

4 Approach

4.1 Overview

Reverse-engineering practice shows that analysts typically form a global view of a binary before examining functions in detail [41, 57],

such as its domain or functional scope. Motivated by this coarse-to-fine reasoning process and by the challenges discussed above, we propose HIERONYM, a function renaming framework that leverages hierarchical code comprehension and multi-source information to robustly handle binaries compiled under diverse optimization levels. HIERONYM consists of three key steps: (I) **Hierarchical Domain Adaptation**, which enhances LLMs’ understanding of binary semantics at multiple granularities; (II) **Multi-Source Information Fusion**, which integrates global binary context, local calling context, and the target function to construct a comprehensive semantic representation; and (III) **Function Renaming Fine-Tuning**, which applies the fused information to the domain-adapted model to enable accurate and generalizable function renaming.

4.2 Hierarchical Domain Adaptation

We build a binary function renaming model upon a pre-trained generative LLM, leveraging its strong semantic modeling capability to capture complex binary function semantics and generalize across diverse binaries. Although general-purpose LLMs are exposed to a limited amount of source code during pre-training (e.g., approximately 5% in the LLaMA dataset [55]), a substantial semantic gap remains between source code and binary code. Source code preserves rich information—such as variable names and types, and structural cues—whereas stripped binaries lose most of these details, rendering direct application of LLMs to such binaries ineffective.

To bridge this gap, we propose a hierarchical summarization-driven domain adaptation strategy, illustrated in Figure 2. We define a code snippet as a function and its constituent statements, where statements correspond to control-flow constructs (e.g., if-else, while, and switch). This strategy employs summaries at two complementary levels to provide a complete and nuanced representation of binary function semantics: function-level summaries capture high-level functional intent, while statement-level summaries model fine-grained implementation logic. Specifically, we extract statements from the abstract syntax tree (AST) and employ an advanced LLM to generate summaries for both statements and entire functions, forming a hierarchical representation that is used to systematically enhance HIERONYM’s understanding of binary semantics.

Statement-Level Data Preparation. Compared with source code, binaries with symbol tables are structurally closer to their stripped

counterparts while preserving richer information, such as identifier names and type details. Leveraging this property, we parse decompiled functions from binaries with symbol tables using Tree-sitter to construct ASTs. From the ASTs, we identify nodes corresponding to control-flow constructs—such as `if-else`, `for`, `while`, `do-while`, and `switch`—as candidate statements, and further decompose nested constructs into multiple independent candidate statements. To improve statement quality, we apply a two-stage refinement process. First, we merge overly short fragments (those containing fewer than five lines of meaningful code) to reduce noise. Second, we merge overlapping fragments whose line-count difference is less than eight lines to eliminate redundancy. The resulting set consists of moderate-sized statements, which serve as the fundamental units for subsequent hierarchical summary generation.

Summary Generation via Code LLMs. Building on the success of code-specialized LLMs (such as Qwen3-Coder [54] and Code Llama [48]) in generating high-quality function summaries, we leverage these models to guide binary code analysis. We do not use original source-code comments, as prior studies show that LLM-generated summaries yield significantly better performance than source-code comments for function renaming [33]. Motivated by these findings, we employ Qwen3-Coder to generate summaries of code snippets from decompiled binaries that retain symbol tables. These summaries provide high-quality semantic supervision for adapting HIERONYM to the binary domain. During summary generation, we guide the LLM to focus on stable features (e.g., inputs/outputs, key steps, and logical structure), ensuring closer alignment with the characteristics of stripped decompiled code.

To improve reliability, we employ self-consistency checking [8, 42, 60]: multiple candidate summaries are generated through repeated LLM queries, the model then evaluates their mutual consistency, and the most consistent summary is selected as the final output. Through experimentation with different prompting strategies, we identify an optimal prompt template.

Alignment of Stripped Binaries and Summaries. To associate summaries derived from DWARF-enabled binaries with stripped binaries, we perform alignment at both the function and statement levels, constructing (stripped function/statement, summary) pairs for training the binary-adapted LLM, as shown in Figure 2. We first conduct function-level alignment by parsing DWARF information and matching stripped functions to their unstripped counterparts based on function addresses (i.e., start and end addresses). As symbol-table stripping does not modify function address, the address-based matching establishes a reliable correspondence, enabling accurate association of summaries with stripped functions.

We then perform statement-level alignment. The stripping process may alter internal function structure, so the line-number-based matching is unreliable. To address this issue, we propose a hybrid alignment strategy that combines structural and semantic similarity to match statements between stripped and unstripped binaries, making the alignment robust to local variations introduced by stripping. Specifically, we parse decompiled code using Tree-sitter to identify logical statements (e.g., `if-else`, `for`, `while`). For each statement, we construct its AST and extract type paths from the root to leaf nodes. Structural similarity is measured using the Jaccard similarity over these AST-path sets. We obtain semantic embeddings for each statement using a pre-trained CodeBERT model [17] and compute

semantic similarity via cosine similarity. The matching score is defined as a weighted combination of structural and semantic similarities. During alignment, we enforce consistency in statement types and accept a match only if its score exceeds a predefined threshold. This procedure enables mapping of stripped statements to summaries generated from their unstripped counterparts.

Based on the above design, we adopt hierarchical summarization as our domain-adaptation strategy. Specifically, we pair decompiled stripped binary code snippets with the corresponding summaries and further fine-tune HIERONYM to adapt it to binary semantics.

4.3 Multi-Source Information Fusion

To acquire task-relevant semantics, we introduce a semantic fusion module that integrates multi-source information to construct a comprehensive representation of function semantics. Specifically, the module incorporates three complementary sources: global binary context, local calling context, and the target function itself. This section describes the representation of each information source.

Global Binary Context. Motivated by reverse engineering practice, we extract domain-relevant information from binaries as global binary context. String literals—which may remain present even in stripped binaries—serve as a primary source of domain-relevant information. These strings are generally stable across compilation settings and often encode domain- or function-related semantic cues. We extract candidate strings by retaining sequences longer than eight characters and applying heuristic filters to remove meaningless patterns. Specifically, we discard low-quality strings (e.g., low-diversity or non-alphanumeric-dominated patterns) and score the remainder in $[0, 1]$ based on alphanumeric ratio, proportion of non-special characters, and penalties for consecutive special characters. Only strings with scores above 0.65 are retained, while system-related strings (e.g., compilation paths) are preserved. From the filtered set, we randomly sample 100 strings and augment them with file-type indicators and dynamic dependency information to construct the final string set. File information provides insights into architecture and build characteristics, while dynamic dependencies that indicate domains (e.g., `libssl/libcurl` suggesting encryption/networking), jointly helping to constrain the binary’s domain. However, string content varies substantially across binaries, and naively concatenating all strings can easily exceed the context-length limits of HIERONYM during fine-tuning. To address this issue, we leverage the summarization capability of an LLM to generate a fixed-length textual summary from the final string set. This summary captures high-level domain context, yielding a compact yet semantically rich representation of global binary context.

Local Calling Context. We construct local calling context based on the direct call relationships of the target function. Since call-path depth varies significantly across functions, incorporating all indirect relationships would produce excessively large call graphs, incurring substantial computational and memory overhead during training and inference. To balance expressiveness and efficiency, we follow prior work [7, 14, 36] and restrict context to direct callers and callees. These functions are categorized as internal or external. As illustrated in Figure 2, the direct context of target function (node 3) includes two callers (nodes 1 and 2) and two callees (nodes 4 and 5), where nodes 1–4 are internal functions and node 5 is external.

We adopt distinct representation strategies for internal and external functions. For internal functions, we use function summaries as their representations. Although decompiled code is available, directly incorporating it presents two challenges: (i) its length may exceed the input limits of HIERONYM, and (ii) long caller or callee code can dilute attention and hinder the model’s focus on the target function. For example, in Figure 1, the decompiled code of the caller `png_read_row` spans 156 lines, whereas the target function contains only 30 lines. Function summaries provide a compact and effective alternative. First, summaries are less sensitive to instruction-level variations introduced by compilation, yielding robust representations. Second, compared with name-propagation approaches [50], which require iterative execution and are prone to error accumulation, summaries convey richer semantics in a concise form and enable parallel processing, improving both efficiency and stability. Finally, summaries reduce input length, better fit the context window of LLMs, and help the model focus on target function semantics. For external functions whose implementations are unavailable (e.g., dynamically linked library functions), we exploit the observation that their semantics can typically be inferred from their names (e.g., Linux system calls [20]). Accordingly, we directly use function names as their representations. The constructed local calling context comprises two components: summaries of internal functions and names of external functions. This design captures both internal implementation semantics and external calling interfaces, yielding a semantically rich and robust representation.

Target Function. We represent the target function using its decompiled code extracted from stripped binaries. Unlike raw assembly, decompiled code preserves richer structural and semantic information, making it more amenable to LLMs. Prior work has shown that using decompiled code as input can improve function renaming performance [33]. However, decompiler-generated code typically contains address-based function names that carry no semantic meaning. Directly incorporating such names can mislead model learning and degrade semantic understanding [33]. Therefore, we preserve the original code structure—including function bodies and parameters—while replacing address-based function names with a special [MASK] token. This strategy removes artificial naming noise and aligns the input representation with the masked-language modeling paradigm commonly used during LLM pretraining, thereby improving compatibility with the model’s learned representations.

4.4 Function Renaming Fine-Tuning

HIERONYM aims to achieve accurate and generalizable function renaming for stripped binaries. To this end, we fine-tune the LLM—already adapted to the binary domain—on a binary function renaming dataset, thereby optimizing its performance for the target task.

Parameter-Efficient Learning. Most existing binary function renaming models (e.g., SymLM [36] and BLens [4]) rely on full-parameter supervised fine-tuning, which updates all model parameters and incurs substantial computational cost. For example, SymLM requires up to 8 days of fine-tuning [36]. Given that HIERONYM is built upon a significantly larger LLM, full-parameter fine-tuning would be prohibitively expensive. To address this issue, we adopt a parameter-efficient fine-tuning strategy based on Low-Rank Adaptation (LoRA) [28]. LoRA freezes the original model parameters and

introduces trainable low-rank matrices, dramatically reducing the number of trainable parameters while often achieving performance comparable to—or better than—full fine-tuning [28, 58].

HIERONYM is built on the LLM with a Transformer decoder architecture, whose core layers include masked multi-head attention, multi-head attention, additive normalization, and feed-forward layers. To efficiently adapt the model to binary semantics, we apply LoRA to the attention layers, which play a central role in semantic representation learning. Specifically, the original attention weight matrices are frozen to preserve the model’s pre-trained language understanding and instruction-following capabilities. During training, gradient updates ($\delta W \in \mathbb{R}^{d \times d}$) are instead captured through a low-rank decomposition into two trainable matrices $W_V \in \mathbb{R}^{d \times r}$ and $W_\Delta \in \mathbb{R}^{r \times d}$. This reduces trainable parameters from $d \times d$ to $2 \times r \times d$ ($r \ll d$), yielding substantial savings in computation and memory. The low-rank updates are incorporated into the original model via residual connections [26], preserving expressive capacity while enabling adaptation to the binary function renaming task.

Fine-Tuning Ground Truth and Objective. Following prior work [11, 18, 33, 36, 69], we use developer-defined function names extracted from binary symbol tables as ground truth during fine-tuning. To match the model’s output space, each ground-truth name is tokenized with the base model’s tokenizer¹.

We formulate binary function renaming as an autoregressive sequence generation task. The input to HIERONYM is denoted as $\mathcal{I}$, which consists of three components: global binary context $\mathcal{B}$, local calling context $\mathcal{C}$, and target function $\mathcal{T}$. The calling context $\mathcal{C}$ includes the callers ($\mathcal{C}^+$) and callees ($\mathcal{C}^-$) of $\mathcal{T}$, i.e., $\mathcal{C} = \{\mathcal{C}^+, \mathcal{C}^-\}$, and the input is $\mathcal{I} = \{\mathcal{B}, \mathcal{C}, \mathcal{T}\}$. Given a tokenized input sequence $\mathcal{I} = \{i_1, i_2, \dots, i_n\}$ and an output sequence $\mathcal{W} = \{w_1, w_2, \dots, w_m\}$, HIERONYM employs a forward autoregressive factorization and is trained to maximize the conditional log-likelihood:

$$\max_{\theta} \log P_{\theta}(\mathcal{W}) = \sum_{t=1}^m \log P_{\theta}(w_t \mid \mathcal{I}; w_{1:t-1}) \quad (2)$$

$$\log P_{\theta}(w_t \mid \mathcal{I}; w_{1:t-1}) = \frac{\exp(\mathbf{h}_{\theta}(\mathcal{I}; w_{1:t-1})^{\top} \mathbf{e}(w_t))}{\sum_{w' \in \mathcal{V}} \exp(\mathbf{h}_{\theta}(\mathcal{I}; w_{1:t-1})^{\top} \mathbf{e}(w'))} \quad (3)$$

where θ denotes the trainable parameters of HIERONYM, $w_{1:t-1}$ represents the sequence of tokens generated prior to predicting w_t , $\mathbf{h}_{\theta}(\cdot)$ is the hidden state vector produced by the intermediate layers of HIERONYM for contextual representation, $\mathbf{e}(\cdot)$ is the corresponding token embedding, and $\mathcal{V}$ is the vocabulary of HIERONYM.

5 Evaluation

In this section, we conduct a comprehensive experimental evaluation to answer the following research questions:

RQ1: How well does HIERONYM rename stripped-binary functions?

RQ2: How does HIERONYM compare with SOTA methods?

RQ3: How does each component affect overall performance?

RQ4: How well does HIERONYM generalize?

5.1 Experimental Setup

Environment. All experiments are run on a server equipped with 48-core Intel Xeon Platinum 8558P CPU, 2 TB of RAM, and 8*NVIDIA

¹We use the Hugging Face tokenizer provided with the base model of HIERONYM.

H100 GPUs, running Ubuntu 22.04 with Python 3.10. We use Ghidra (version 11.2.1) to extract functions from binaries and generate decompiled code. Although HIERONYM is decompiler-agnostic, Ghidra is selected due to its open-source availability and widespread adoption in prior research. To ensure fair comparison, we adopt Code Llama model as the base model, consistent with the SOTA method SymGen [33]. Code Llama is a code-specialized variant of Llama 2, pretrained on six programming languages [48]. Parsing of decompiled code is implemented using ANTLR 4 [43]. Parameter-efficient fine-tuning is conducted using PyTorch, the Hugging Face Transformers library [62], and LlamaFactory [72]. For function-name preprocessing and evaluation, we use NLTK [5], python-Levenshtein [23], and the Qwen3-Embedding model [71].

Baselines. In this paper, we select SymLM [36], XFL [46], SymGen [33], and BLens [4] as baselines for comparison. All baselines are trained on our dataset under the same experimental settings as HIERONYM to ensure a fair comparison, and we report results from the best-performing hyperparameter configurations following each method’s training guidelines. XFL is applicable only to x64 binaries and BLens builds upon XFL’s representations, both XFL and BLens are evaluated exclusively on the x64 dataset and compared against HIERONYM. We exclude NERO [11], NFRE [18], or Epitome [69] from comparison, as prior studies have shown that the selected baselines consistently outperform them. We also omit llasm [50], as its full source code is unavailable and its results could not be reproduced despite attempts to contact the authors.

5.2 Dataset Construction

We evaluate HIERONYM on the dataset introduced by SymGen [33], comprising 33 open-source projects and their corresponding binaries compiled with GCC 9.4.0 across four optimization levels (O0–O3) and four architectures (x86, x64, ARM, and MIPS). The selected projects originate from widely used libraries, which are extensively employed in prior function renaming studies [18, 36, 69]. **Function Renaming Dataset.** The dataset contains 9,842 unique binaries and 2,237,915 binary functions in total. Unlike prior methods such as SymGen and SymLM, which train separate models for each architecture–optimization pair, HIERONYM trains a single model per architecture, thereby reducing training overhead. Following prior work [33, 36, 69], we randomly split the dataset into training, validation, and test sets in an 8:1:1 ratio.

To prevent data leakage [24], we split the dataset based on source code: binaries derived from the same source file but compiled with different optimization levels are assigned to the same sets. Although function duplication due to code reuse and shared libraries is common in real-world binaries, we follow the stricter deduplication strategy of SymGen to evaluate generalization. Specifically, we remove functions that share the same name, have identical bodies despite different names, or originate from the same source but differ only due to address-dependent callees. Consistent with previous studies [4, 46], we exclude empty functions, functions with meaningless names, and functions automatically generated by Ghidra.

Since HIERONYM adopts the same base model and dataset configuration as SymGen, whose study demonstrated that the dataset was not leaked into the pretraining corpus of Code Llama [48],

potential data-contamination concerns are mitigated, ensuring a fair comparison with SOTA approaches.

Hierarchical Domain Adaptation Dataset. Following the procedure described in Section 4.2, we construct the domain-adaptation dataset exclusively from the training set. Specifically, we first generate summaries for code snippets from decompiled code with symbol tables using a code-specialized LLM, and then align these summaries with the corresponding stripped decompiled code snippets.

5.3 RQ1: Overall Effectiveness

Our analysis begins with a comprehensive evaluation of HIERONYM’s performance. Token-level results across four architectures and four optimization levels are reported in Table 1, while name-level results across the four architectures are summarized in Table 2. Overall, HIERONYM demonstrates strong performance, though clear variations emerge across different architectures. At the token level, HIERONYM achieves weighted-macro precision, recall, and F1-score of 0.4487, 0.4223, and 0.4339, respectively. When examined by architecture, HIERONYM performs best on x64 binaries, attaining an average precision of 0.5186, a recall of 0.5329, and an F1-score of 0.5256. Results on x86 binaries are slightly lower, with a precision of 0.5063, a recall of 0.4630, and an F1-score of 0.4832. In contrast, performance declines on ARM and MIPS binaries: HIERONYM attains a precision of 0.4258, recall of 0.4220, and F1-score of 0.4239 for ARM, and the corresponding values are 0.3440, 0.2713, and 0.3031 for MIPS. In the name-level evaluation, the performance trend is largely consistent with that observed at the token level, although minor deviations arise on the ARM architecture. Specifically, HIERONYM achieves the highest accuracy on x64 (0.4254), followed by ARM (0.3455) and x86 (0.3220), with performance dropping markedly on MIPS (0.1709). As name-level evaluation is conducted on a sampled subset of the test set, these results may be affected by the dataset scale. Nevertheless, the overall trends remain consistent with token-level findings, indicating stable model behavior across architectures.

The observed performance gap between x86/x64 and ARM/MIPS architectures is consistent with that of baselines under identical settings. For example, HIERONYM achieves approximately 1.24× higher performance on x64 than on ARM, comparable to the 1.38× improvement reported for SymGen. We attribute this disparity primarily to architecture-specific factors, particularly the greater complexity of application binary interfaces in ARM and MIPS. Their calling conventions and register usage patterns introduce additional challenges for function renaming. Moreover, empirical analysis shows that ARM binaries compiled from the same source often yield longer and more structurally complex decompiled functions than their x64 counterparts, with a higher incidence of variable-type recovery errors. These architecture-dependent characteristics, combined with errors introduced during the decompilation process, exacerbate the difficulty of semantic modeling, thereby contributing to the reduced performance of HIERONYM on ARM and MIPS architectures.

In addition, we conduct a manual evaluation on 1,600 predictions sampled from the C experiments to assess the reliability of our evaluation framework. The results show that token-level semantic alignment achieves 89.8% agreement with human judgment, while name-level evaluation reaches 89.2%. Moreover, the name-level metric reduces GPT hallucinations by 11.28%. These findings

Table 1: Overall Token-Level Performance of HIERONYM and Our Baselines. Note that $\uparrow$ indicates the improvement of HIERONYM over the corresponding baseline.

Architecture	Model	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
		O0			O1			O2			O3		
x64	HIERONYM	0.5262	0.5620	0.5435	0.5373	0.5342	0.5357	0.4991	0.5125	0.5057	0.5117	0.5230	0.5173
	SymGen	0.4103	0.3933	0.4016	0.4180	0.3953	0.4064	0.4158	0.3750	0.3943	0.4044	0.4000	0.4022
	BLens	0.1416	0.1031	0.1193	0.1397	0.1092	0.1225	0.0926	0.0602	0.0730	0.1188	0.0755	0.0923
	XFL	0.1041	0.0788	0.0897	0.1712	0.1126	0.1359	0.1000	0.0721	0.0838	0.1122	0.0680	0.0847
	SymLM	0.0376	0.0854	0.0522	0.0979	0.0550	0.0705	0.0749	0.0462	0.0571	0.0922	0.0569	0.0704
	HIERONYM vs SymGen	($\uparrow 28.24\%$)	($\uparrow 42.90\%$)	($\uparrow 35.34\%$)	($\uparrow 28.54\%$)	($\uparrow 35.14\%$)	($\uparrow 31.83\%$)	($\uparrow 20.03\%$)	($\uparrow 36.67\%$)	($\uparrow 28.25\%$)	($\uparrow 26.53\%$)	($\uparrow 30.74\%$)	($\uparrow 28.61\%$)
	HIERONYM vs BLens	($\uparrow 271.60\%$)	($\uparrow 445.12\%$)	($\uparrow 355.59\%$)	($\uparrow 284.62\%$)	($\uparrow 389.19\%$)	($\uparrow 337.35\%$)	($\uparrow 438.96\%$)	($\uparrow 751.35\%$)	($\uparrow 592.75\%$)	($\uparrow 330.71\%$)	($\uparrow 592.68\%$)	($\uparrow 460.42\%$)
	HIERONYM vs XFL	($\uparrow 405.46\%$)	($\uparrow 613.23\%$)	($\uparrow 505.93\%$)	($\uparrow 213.85\%$)	($\uparrow 374.42\%$)	($\uparrow 294.22\%$)	($\uparrow 399.08\%$)	($\uparrow 610.84\%$)	($\uparrow 503.47\%$)	($\uparrow 356.04\%$)	($\uparrow 669.08\%$)	($\uparrow 510.70\%$)
x86	HIERONYM vs SymLM	($\uparrow 1299.44\%$)	($\uparrow 558.11\%$)	($\uparrow 941.22\%$)	($\uparrow 448.83\%$)	($\uparrow 871.27\%$)	($\uparrow 659.93\%$)	($\uparrow 566.32\%$)	($\uparrow 1009.34\%$)	($\uparrow 785.65\%$)	($\uparrow 454.97\%$)	($\uparrow 819.11\%$)	($\uparrow 634.75\%$)
	HIERONYM	0.5016	0.4085	0.4503	0.5057	0.4671	0.4856	0.4981	0.4700	0.4836	0.5200	0.5065	0.5131
	SymGen	0.3656	0.3243	0.3437	0.4049	0.3468	0.3736	0.3825	0.3436	0.3620	0.3981	0.3766	0.3871
	SymLM	0.1518	0.0660	0.0920	0.1385	0.0674	0.0907	0.1017	0.0736	0.0854	0.1320	0.0903	0.1073
	HIERONYM vs SymGen	($\uparrow 37.19\%$)	($\uparrow 25.96\%$)	($\uparrow 31.00\%$)	($\uparrow 24.89\%$)	($\uparrow 34.68\%$)	($\uparrow 29.98\%$)	($\uparrow 30.23\%$)	($\uparrow 36.78\%$)	($\uparrow 33.60\%$)	($\uparrow 30.61\%$)	($\uparrow 34.49\%$)	($\uparrow 32.56\%$)
	HIERONYM vs SymLM	($\uparrow 230.41\%$)	($\uparrow 518.90\%$)	($\uparrow 389.41\%$)	($\uparrow 265.12\%$)	($\uparrow 592.96\%$)	($\uparrow 435.40\%$)	($\uparrow 389.79\%$)	($\uparrow 538.54\%$)	($\uparrow 466.31\%$)	($\uparrow 293.91\%$)	($\uparrow 460.89\%$)	($\uparrow 378.22\%$)
	HIERONYM	0.3862	0.3835	0.3848	0.4584	0.4542	0.4563	0.4530	0.4489	0.4510	0.4058	0.4015	0.4036
	SymGen	0.3144	0.3054	0.3098	0.2807	0.2544	0.2669	0.3192	0.3022	0.3105	0.2879	0.2670	0.2771
ARM	SymLM	0.0587	0.0263	0.0363	0.0472	0.0349	0.0401	0.0645	0.0394	0.0489	0.0496	0.0213	0.0298
	HIERONYM vs SymGen	($\uparrow 22.84\%$)	($\uparrow 25.56\%$)	($\uparrow 24.22\%$)	($\uparrow 63.29\%$)	($\uparrow 78.54\%$)	($\uparrow 70.96\%$)	($\uparrow 41.92\%$)	($\uparrow 48.55\%$)	($\uparrow 45.24\%$)	($\uparrow 40.95\%$)	($\uparrow 50.37\%$)	($\uparrow 45.66\%$)
	HIERONYM vs SymLM	($\uparrow 557.93\%$)	($\uparrow 1358.06\%$)	($\uparrow 960.15\%$)	($\uparrow 871.12\%$)	($\uparrow 1201.46\%$)	($\uparrow 1037.85\%$)	($\uparrow 602.33\%$)	($\uparrow 1039.40\%$)	($\uparrow 822.20\%$)	($\uparrow 718.11\%$)	($\uparrow 1784.86\%$)	($\uparrow 1254.42\%$)
	HIERONYM	0.3543	0.2650	0.3032	0.3630	0.2891	0.3218	0.3684	0.2809	0.3188	0.2902	0.2502	0.2687
	SymGen	0.1881	0.2029	0.1952	0.1812	0.2074	0.1935	0.1822	0.1787	0.1804	0.1345	0.1560	0.1445
	SymLM	0.0198	0.0097	0.0130	0.0256	0.0095	0.0139	0.0185	0.0070	0.0102	0.0128	0.0046	0.0068
	HIERONYM vs SymGen	($\uparrow 88.38\%$)	($\uparrow 30.61\%$)	($\uparrow 55.34\%$)	($\uparrow 100.31\%$)	($\uparrow 39.39\%$)	($\uparrow 66.33\%$)	($\uparrow 102.21\%$)	($\uparrow 57.21\%$)	($\uparrow 76.71\%$)	($\uparrow 115.75\%$)	($\uparrow 60.40\%$)	($\uparrow 85.97\%$)
	HIERONYM vs SymLM	($\uparrow 1689.62\%$)	($\uparrow 2631.95\%$)	($\uparrow 2232.52\%$)	($\uparrow 1317.81\%$)	($\uparrow 2943.21\%$)	($\uparrow 2215.46\%$)	($\uparrow 1891.52\%$)	($\uparrow 3913.38\%$)	($\uparrow 3025.39\%$)	($\uparrow 2167.03\%$)	($\uparrow 5339.78\%$)	($\uparrow 3851.88\%$)

*XFL targets only x64 binaries by design [46], and BLens [4] builds upon XFL. Accordingly, both are evaluated solely on the x64 dataset.

Table 2: Name-Level Accuracy of HIERONYM and SymGen.

Model	Architecture			
	x64	x86	ARM	MIPS
HIERONYM	0.4254	0.3220	0.3455	0.1709
SymGen	0.3331	0.2251	0.1530	0.0765

demonstrate that our evaluation framework is both reliable and closely aligned with human judgment.

5.4 RQ2: Baseline Comparison

To further deepen our analysis, we compare HIERONYM with baselines on the same dataset. Token-level and name-level evaluation results are reported in Table 1 and Table 2, respectively.

Token-Level Comparison. Overall, HIERONYM substantially outperforms all baselines, achieving improvements of up to 396.37%, 688.01%, and 547.72% in precision, recall, and F1-score, respectively. Traditional ML-based baselines exhibit notably weaker token-level performance on unseen functions. Among the x64-only methods, BLens achieves an average precision of 0.1232, recall of 0.0870, and F1-score of 0.1018, while XFL reports comparable results (precision 0.1219, recall 0.0829, F1-score 0.0985). SymLM performs the weakest overall, with average precision, recall, and F1-score of 0.0702, 0.0433, and 0.0515, respectively, across all architectures and optimization levels. In contrast, HIERONYM consistently surpasses these baselines by a wide margin. Compared with SymLM, HIERONYM achieves improvements of 860.27% in precision, 1598.83% in recall, and 1255.60% in F1-score. Although XFL outperforms SymLM on x64 binaries, HIERONYM still exceeds XFL by 343.61% in precision, 566.89% in recall, and 453.58% in F1-score. Similarly, relative to BLens, HIERONYM delivers achieves gains of 331.47%, 544.59%, and 436.52% in precision, recall, and F1-score, respectively. Notably, the performance of ML-based baselines on our dataset is substantially lower than that

reported in their studies. For example, SymLM reports an average F1-score of 0.73 on its dataset but attains 0.0515 in our evaluation. This discrepancy is primarily attributable to data leakage in prior experimental setups, where duplicated functions between training and test sets inflate performance and obscure true generalization. Consistent with this explanation, SymLM reports F1-score below 0.10 on unseen binaries in its study [36]. Similar degradation on unseen functions has also been reported for XFL [46] and BLens [4]. Moreover, baseline methods that incorporate calling information are highly sensitive to optimizations: SymLM’s F1-score varies by 35.06%, XFL by 62.17%, and BLens by 67.81% across optimization levels. HIERONYM exhibits only a 7.48% variation, demonstrating strong robustness to optimization level changes. A similar trend appears across architectures. SymLM shows F1-score variation up to 753.64%, while HIERONYM exhibits substantially smaller variation (73.41%), highlighting robustness to architectural differences.

The LLM-based baseline SymGen achieves average token-level precision, recall, and F1-score of 0.3180, 0.3018, and 0.3093 across all architectures and optimization levels. Since we adopt the same configuration as in the original SymGen study [33], these results are not far from those previously reported. Even under this strong and directly comparable baseline, HIERONYM demonstrates a clear advantage, improving precision by 50.12%, recall by 41.75%, and F1-score by 45.10%. To further ensure fairness, we also re-evaluate performance using SymGen’s evaluation metrics, under which HIERONYM still improves F1-score by 47.96% over SymGen. These results indicate that LLM-based methods generalize more effectively to unseen binaries than traditional ML methods, and that the multi-source information and hierarchical summarization-driven domain adaptation employed by HIERONYM further enhance semantic understanding and naming performance.

Name-Level Comparison. Due to budget constraints and the fact that ML-based methods are restricted to predicting labels from a

closed vocabulary, we perform name-level evaluation only for HIERONYM and SymGen, as SymGen substantially outperforms other baselines at the token level. Under this setting, HIERONYM achieves an average accuracy improvement of 79.94% over SymGen across all architectures. This result further demonstrates that HIERONYM captures function semantics more comprehensively and generates function names with higher semantic accuracy.

5.5 RQ3: Ablation Study

In this section, we conduct ablation studies to obtain deeper insights into HIERONYM. Our goal is to systematically evaluate the contribution of each individual component to the overall performance. To ensure a controlled and consistent comparison, all ablation experiments are performed on x64 dataset and use token-level metrics.

Effectiveness of Hierarchical Domain Adaptation. Domain adaptation is designed to enhance HIERONYM’s ability to capture binary code semantics by bridging the knowledge gap between the general-purpose pre-trained LLM and the binary domain. To evaluate the effectiveness of this strategy, we compare three model configurations: (1) hierarchical summarization-based domain adaptation, (2) function summarization only domain adaptation, and (3) no domain adaptation, where the model is directly fine-tuned on the function-renaming task using parameter-efficient methods. This comparison allows us to isolate and quantify the contribution of hierarchical summarization-based domain adaptation.

The results are shown in Figure 3. Across the four optimization levels (O0–O3) on the x64 architecture, hierarchical summarization-based domain adaptation improves HIERONYM over the no-adaptation baseline by an average of 11.14%, 26.76%, and 18.83% in precision, recall, and F1-score, respectively. Compared with an LLM adapted using function summarization only adaptation, HIERONYM yields average gains of 8.49%, 9.43%, and 8.95% on the same metrics. Function summarization only adaptation also outperforms the no-adaptation configuration, achieving average improvements of 2.44%, 15.85%, and 9.06% in precision, recall, and F1-score. These results lead to two key observations. First, domain adaptation enhances the LLM’s ability to understand binary code semantics. Second, incorporating hierarchical summarization provides additional benefits, demonstrating that fine-grained, statement-level semantic learning effectively complements function-level understanding and plays an important role in improving function renaming performance.

Effectiveness of Global Binary Context. To assess the impact of global binary context, we compare fine-tuned variants of HIERONYM with and without incorporating this context as input. As shown in Figure 4, incorporating global context yields a consistent performance gain, increasing the average F1-score from 0.4117 to 0.5256 (a relative improvement of 27.63%). This demonstrates the critical role of global context in function renaming, as it provides domain-relevant information that enhances the model’s ability to capture function semantics and generate accurate names.

To further verify that these gains are not driven by binary names that may appear in the global context, we remove test samples whose context contains such information and re-evaluate on the filtered set. HIERONYM achieves an F1-score of 0.444 on this subset, compared to 0.4339 on the original test set. The stable performance

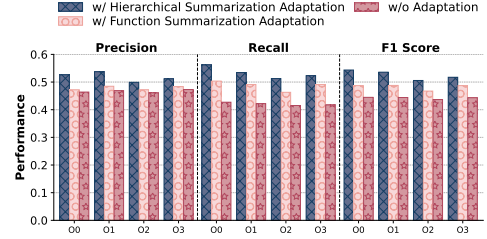


Figure 3: Effectiveness of Hierarchical Domain Adaptation

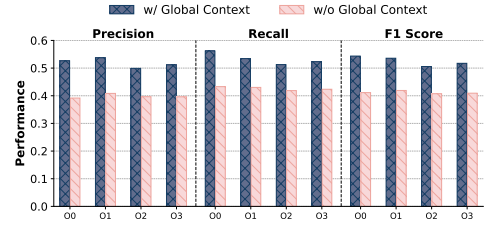


Figure 4: Effectiveness of Global Binary Context

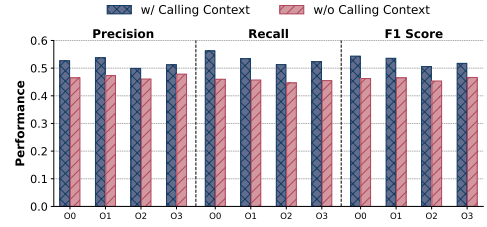


Figure 5: Effectiveness of Calling Context

indicates that the improvements primarily arise from effective use of domain-relevant context rather than leakage from binary names.

Effectiveness of Calling Context. To evaluate the contribution of calling-context information, we compare two fine-tuned variants of HIERONYM: one that incorporates calling context in the input and another that excludes it. As shown in Figure 5, incorporating calling context consistently improves performance, yielding average gains of 10.52% in precision, 17.22% in recall, and 13.81% in F1-score. Notably, these improvements are more pronounced at lower optimization levels (O0/O1) and diminish at higher levels (O2/O3). We attribute this trend to the effects of aggressive compiler optimizations, such as function inlining, which remove explicit inter-function call relationships, thereby reducing the effectiveness of calling context. Overall, these results demonstrate that calling context provides valuable complementary semantic cues, enabling HIERONYM to construct richer function semantics representations and improve function renaming performance.

5.6 RQ4: Generalizability Test

In this section, we conduct a comprehensive evaluation of HIERONYM’s generalization across multiple dimensions, including

Table 3: Cross-Compiler Performance

Architecture	HIERONYM			SymGen		
	Precision	Recall	F1-score	Precision	Recall	F1-score
x64	0.5841	0.6240	0.6034	0.4447	0.3950	0.4178
x86	0.5462	0.4916	0.5175	0.4253	0.3780	0.4001
ARM	0.3848	0.3442	0.3633	0.2708	0.2346	0.2512
MIPS	0.3626	0.3396	0.3507	0.1966	0.2038	0.2000

Table 4: Cross-Compiler Performance

Architecture	HIERONYM			SymGen		
	Precision	Recall	F1-score	Precision	Recall	F1-score
x64	0.4817	0.4793	0.4805	0.4238	0.3394	0.3769
x86	0.4849	0.3994	0.4379	0.4451	0.3526	0.3931
ARM	0.5049	0.4427	0.4717	0.4167	0.3573	0.3846
MIPS	0.4753	0.4518	0.4632	0.4020	0.3736	0.3866

Table 5: Performance on Obfuscated Binaries

Obfuscation	HIERONYM	SymGen
w/o obfuscation	0.4095	0.2947
bcfobf	0.3866 (-5.58%)	0.2759 (-6.37%)
cfobf	0.3759 (-8.20%)	0.2679 (-9.12%)
subobf	0.303 (-0.17%)	0.2911 (-1.22%)

cross-compiler, cross-decompiler, cross-project, and obfuscated settings. We use SymGen as the sole baseline, as it achieves the best performance in Section 5.4. To ensure fair comparison, all experiments adopt the same test set split and strict deduplication strategy as in Section 5.2, and are evaluated using token-level metrics.

Cross-Compiler Evaluation. We recompile the 33 projects in SymGen dataset using Clang 14.0.0 and re-evaluate both HIERONYM and SymGen on the resulting binaries. As shown in Table 3, HIERONYM achieves a weighted macro F1-score of 0.4587, compared to 0.3173 for SymGen, corresponding to a 49.68% improvement. On the GCC-compiled test set, the F1-scores are 0.4339 for HIERONYM and 0.3093 for SymGen. The comparable performance across compilers indicates strong cross-compiler generalization for both methods, likely because LLM-based methods operate on decompiled code, which mitigates compiler-specific variation. Notably, HIERONYM further benefits from its design, leading to improved naming performance.

Cross-Compiler Evaluation. We re-decompile the test set using IDA Pro 9.0 and re-evaluate both HIERONYM and SymGen on the resulting dataset. As shown in Table 4, HIERONYM achieves a weighted macro-average F1-score of 0.4633, compared to 0.3853 for SymGen, yielding a 20.54% improvement. Compared to Ghidra, IDA can recover additional referenced string information—particularly for ARM and MIPS binaries—which can improve renaming performance. Consequently, both methods benefit from this change, with SymGen showing a more pronounced gain. Nevertheless, HIERONYM consistently outperforms SymGen, demonstrating robust cross-decompiler generalization.

Cross-Project Evaluation. We further deduplicate the test set by removing samples from projects present in the training set and re-evaluate on the resulting dataset. HIERONYM achieves a weighted

macro-average precision of 0.4897, recall of 0.4461, and F1-score of 0.4661, while SymGen attains 0.3453 precision, 0.3318 recall, and 0.3377 F1-score. These results are comparable to those on the original test set, indicating that both methods maintain strong performance on samples from previously unseen projects and exhibit robust cross-project generalization. This also confirms that our deduplication strategy is both strict and effective.

Obfuscated Binary Evaluation. To evaluate the obfuscation resistance performance of HIERONYM, we use SymGen’s obfuscated binary dataset, which contains 363 binaries from six GNU projects (findutils, coreutils, curl, less, putty, and bash). The binaries are obfuscated using three techniques: bogus control flow (bcfobf), control-flow flattening (cfobf), and instruction substitution (subobf). We also include the original binaries (w/o obfuscation) for comparison. As shown in Table 5, HIERONYM achieves F1-scores of 0.3866, 0.3759, and 0.303 across the three obfuscation settings, outperforming SymGen by up to 40.42%. Relative to original binaries, HIERONYM exhibits only modest degradation (e.g., 0.17% on subobf), which is lower than that of SymGen. These results demonstrate the superior obfuscation robustness of HIERONYM, likely due to its generalizable learning paradigm.

Cross-Language Evaluation. We evaluate the transferability of HIERONYM on C++ binaries and Java code. First, we assess HIERONYM on the Xalan benchmark (x86-64 architecture) from SPEC CPU2017 [10], which contains numerous functions with object-oriented C++ characteristics. HIERONYM achieves an F1-score of 0.4093, compared to 0.3696 for SymGen. Although both methods perform worse than on the original x64 dataset (0.5256 and 0.4011, respectively), HIERONYM still surpasses SymGen by 10.71%. Next, we randomly select four Java projects from AndroZooOpen [40] and apply R8 [12] to obfuscate identifier names, yielding more than 4,400 functions. We then predict only method names. SymGen achieves an F1-score of 0.2977, whereas HIERONYM attains 0.3779, representing a 26.93% improvement. Compared with C++, performance degrades more substantially on Java, likely due to the larger semantic or structural differences between Java and C-family binaries, which pose additional challenges for generalization. Nevertheless, HIERONYM consistently outperforms prior methods across languages, demonstrating strong cross-language transferability.

6 Illustration

6.1 Qualitative Evaluation

In our qualitative analysis, we examine function names predicted by HIERONYM and SymGen, sampled from the inference results. Table 6 reports both ground-truth and predictions, tokenized using our tokenization method and SymGen’s tokenization method, respectively. Tokenization is necessary for token-level evaluation.

Several insightful observations can be drawn from these examples from projects such as poke, binutils, gettext, libpng, freeipmi, and openssl. First, HIERONYM accurately infers function names that include project-specific tokens (e.g., jitter_mangle). As discussed earlier, although SymGen captures the function’s semantics, it fails to recover such identifiers (e.g., jitter), highlighting the effectiveness of our multi-source information fusion strategy. Second, HIERONYM correctly infers names (e.g., ctf_type_encoding) where

Table 6: Qualitative Examples of Predictions Made by HIERONYM and SymGen. Tok. (Ours) denotes our tokenization, and Tok. (SymGen) denotes SymGen’s tokenization.

Ground Truth			HIERONYM Prediction			SymGen Prediction		
Raw	Tok. (Ours)	Tok. (SymGen)	Raw	Tok. (Ours)	Tok. (SymGen)	Raw	Tok. (Ours)	Tok. (SymGen)
jitterc_mangle collector_gettid	jitter,c,mangle collector,get, thread,id	jitter,c,mangle collector,get,t,id	jitter_mangle collector_get_thread_ id	jitter,mangle collector,get,thread,id	jitter,mangle collector,get,thread, id	mangle_name get_thread_id	mangle,name get,thread,id	mangle,name get,thread,id
ctf_type_encoding check_message	ctf,type,encoding check,message	ct,f,type,encode check,message	ctf_type_encoding validate_message	ctf,type,encoding validate,message	ct,f,type,encode validate,message	get_size nc_mvcur	get,size nc,mv,cur	get,size nc,mvc,ur
is_ascii_string	is,ascii,string	be,ascii,string	libgettextpo_is_ascii_ string	lib,get,text,po,is,ascii, string	lib,get,text,po,is, ascii,string	c_isascii_ string	c,is,ascii,string	c,isa,sc,ii,string
png_set_IHDR	png,set,i,hdr	png,set,i,h,d,r	libtextstyle_png_set_ ihdr	lib,text,style,png,set, ihdr	lib,text,style,png, set,i,h,d,r	set_field_type	set,field,type	set,field,type
linearization_ string	linearization, string	linear,ization,string	sensor_reading_type_ code_linearization	sensor,reading,type, code.linearization	sensor,read,type, code.linear,zation	gnuplot_ scale_type	gnuplot,scale, type	gnuplot,scale, type
test_standard_ methods	test,standard, methods	t,est,standard,method	test_asn1_standard_ methods	test,asn,standard, methods	t,est,asn,standard, method	test_pkey_id_ order	test,pkey,id, order	t,est,pkey,id, order

SymGen fails to capture the correct semantics and produces erroneous predictions. Even when the predictions differ lexically from ground truth, HIERONYM often generates semantically equivalent tokens (e.g., `check`, `validate`), demonstrating a deeper understanding of function semantics. In contrast, SymGen frequently fails to grasp the underlying semantics, yielding less meaningful names.

We also observe a form of inference bias in HIERONYM’s predictions, where additional tokens appear that are absent from the ground truth. This phenomenon primarily stems from two sources. (i) Global binary context may include binary names or related library identifiers, which can be incorporated into predictions—for example, tokens such as `libgettextpo` and `libtextstyle` reflect domain cues present in the global context. (ii) Calling context may introduce keywords relevant to function behavior but not part of the canonical name. For instance, when predicting `linearization_string`, HIERONYM introduces tokens such as `sensor` and `type` derived from its calling context. Despite these extraneous tokens, the predicted names consistently capture the core functionality of the target functions, indicating that HIERONYM remains robust in modeling essential function semantics.

Based on the tokenized outputs, we further analyze the differences between our tokenization strategy and SymGen’s, as well as their impact on token-level evaluation results. SymGen often produces overly fine-grained and sometimes erroneous segmentations (e.g., tokenizing ‘`test`’ into {`t`, `est`}, ‘`ctf`’ into {`ct`, `f`}, or decomposing ‘`isascii`’ into {`isa`, `sc`, `ii`}). In contrast, our approach aims to partition function names into semantically meaningful tokens: ‘`isascii`’ is correctly tokenized as {`is`, `ascii`}, while common abbreviations such as ‘`ctf`’ and frequent words like ‘`test`’ are retained as single tokens to avoid unnecessary fragmentation. These differences have a direct and substantial impact on token-level evaluation. Improper tokenization disrupts semantic coherence of tokens and introduces bias into metric computation. For example, for the function name ‘`c_isascii_string`’, the F1-score under SymGen’s tokenization is only 0.25, whereas it increases to 0.857 when using our tokenization strategy. This improvement highlights the effectiveness of our approach in preserving semantic integrity during evaluation. Furthermore, we also conduct name-level evaluation, which is inherently insensitive to tokenization and provides a robust evaluation of the performance of function-renaming methods.

6.2 Effectiveness on Real-world Binaries

To evaluate HIERONYM’s generalization in real-world security analysis, we apply it to the task of function renaming on malware executables. We collect ELF-format malware samples from VirusShare [56] and retain only those containing symbol tables to ensure reliable ground-truth function names. We select various types of x64 Linux malware, such as CDorked, camelot, and DnsAmp. All symbol tables are stripped prior to evaluation, forming an external test set.

Without additional fine-tuning, we directly apply HIERONYM trained on our primary dataset—composed mainly of GNU projects—to rename 2,478 malware functions. Evaluation follows token-level metrics described in Section 3.1, comparing predicted names against ground truth extracted from the original symbol tables. HIERONYM achieves a precision of 0.4855, a recall of 0.4897, and an F1-score of 0.4876. Notably, despite substantial differences in functional intent between benign open-source software and malware, performance on malware binaries is comparable to that observed on our evaluation dataset. These results indicate that HIERONYM retains robust semantic modeling and name inference capability when applied to previously unseen and domain-divergent binaries, highlighting its strong generalization potential in practical security analysis.

7 Limitation & Future Work

Base Model. To ensure fair experimental comparison, we adopt the same base model and experimental setup as the SOTA method SymGen [33]. However, the landscape of generative LLMs is evolving rapidly, with increasingly capable models continually emerging. As HIERONYM is model-agnostic by design, its performance could be further improved by integrating larger or more advanced base models. While our experiments already demonstrate the effectiveness of the proposed approach, investigating its integration with alternative base models represents a promising direction.

Statement Extraction Effectiveness. During dataset construction, we employ heuristic rules to extract code statements from functions. The lack of objective metrics for assessing statement quality introduces potential limitations and leaves room for further refinement. Despite this, our experimental results demonstrate that statement-level summarization effectively supports binary function renaming. Future work will focus on improving the statement extraction algorithm to further enhance performance.

Dataset Quality. Dataset quality is a critical factor in effective model training. For the function renaming task, we employ Ghidra to generate decompiled code as model input. Nevertheless, decompiler outputs inevitably contain inaccuracies due to the inherent challenges of decompilation, and discrepancies across different decompilers may introduce systematic bias. Although HIERONYM is designed to be decompiler-agnostic, its performance could benefit from higher-fidelity decompilation results. In addition, the quality of function names directly affects dataset validity. Many non-descriptive or semantically ambiguous names remain prevalent in real-world binaries. Fundamentally addressing this noise typically requires expert-driven manual curation, which is costly and time-consuming. Developing automated techniques to filter low-quality function names represents an important direction for future work.

For the domain-adaptation dataset, we employ Qwen3-Coder-30B to generate natural-language summaries of code snippets, which serve as supervision for training HIERONYM. The quality and stylistic consistency of these summaries may vary depending on the LLM used. Although proprietary models (e.g., ChatGPT) often produce higher-quality outputs, their cost renders them impractical for large-scale dataset construction. Prior work [33] has shown that summaries generated by open-source LLMs can still effectively support function renaming task. We therefore expect that further improving the consistency of code-snippet summaries could yield additional performance gains for HIERONYM. Exploring this direction constitutes a promising direction for future research.

Evaluator Selection. Using local LLMs similar to those in HIERONYM as evaluators may introduce bias. We therefore adopt GPT as a widely used external judge. We further evaluate Qwen3-Code-30B on 1,600 samples from the C experiments as an alternative evaluator. It achieves 76.31% agreement with human judgments, substantially lower than GPT’s 87.56%. Developing open-source evaluators that match GPT’s performance is an important direction.

Inference Efficiency and Cost. HIERONYM requires an average of 20.69 tokens and 0.0333s per function name, compared to 14.86 tokens and 0.0245s for SymGen. HIERONYM’s total cost comprises global-context summarization (0.0371s), calling-context summarization (0.0351s), and name generation (0.0333s), with the first two stages parallelizable. On large binaries (2,284 functions), HIERONYM achieves a throughput of 3.27 functions/s, corresponding to an end-to-end inference latency of 0.3056s per function.

8 Related Work

8.1 Function Renaming For Stripped Binaries

Early research on binary function renaming primarily relied on traditional ML techniques. For example, Debin [25] transforms binary code into an intermediate representation and constructs variable dependency graphs, using Conditional Random Fields (CRFs) to infer function names. Similarly, Punstrip [45] generates probabilistic binary fingerprints from engineered features and applies a CRF-based model for name prediction. XFL [46] formulates function renaming as a multi-label classification problem that predicts sets of words. More recent works have framed function renaming as a neural machine translation task, translating binary code into natural-language function names. NERO [11] represents calling context using augmented control-flow graphs (CFGs) derived from

assembly instructions, but largely ignores non-call instructions, which are unavailable for many functions. NFRE [18] proposes structure-sensitive embeddings of assembly instructions to generate function names, while AsmDepictor [37] employs a Transformer-based model to encode function semantics from assembly code. SymLM [36] leverages a pretrained assembly-language model to capture execution behavior, and Epitome [69] combines a pretrained assembly-language model with graph neural networks over CFGs to learn function semantics. BLens [4] adopts a different perspective by casting function renaming as an image-captioning-like task, aggregating multiple assembly-based embeddings into an ensemble representation aligned with the latent space of function names. Despite their methodological differences, these approaches share a fundamental limitation: function names are typically generated by selecting tokens from a closed training vocabulary, which restricts generalization to unseen or semantically novel names. Recently, LLMs have demonstrated strong generalization capabilities for function renaming. llasm [50] integrates a pretrained assembly encoder with a decoder-only LLM for name generation. SymGen [33] uses decompiled code as its input and introduces a function-summarization-based adaptation strategy to guide name generation. In this work, we discuss the limitations of existing methods (Section 2.2) and compare HIERONYM with four SOTA methods—SymLM, XFL, BLens, and SymGen—in Section 5.4.

In addition, with the rapid advancement of LLMs, researchers have explored their applications to other stripped binary analysis tasks, including decompilation [39, 63], variable name recovery [64, 66], and variable type recovery [15, 61, 64]. LLM-based approaches introduce a new paradigm for stripped binary analysis and continue to drive progress in the field. These efforts complement our work and further advance research on stripped binary analysis.

8.2 Source Code Summarization

Source code summarization aims to generate concise and human-readable natural-language descriptions for code snippets. Since around 2016, the majority of approaches in this area have been data-driven [32]. One of the earliest seminal works [32] introduced an attention-based encoder-decoder architecture inspired by neural machine translation. Building on this paradigm, CodeBERT [17], a Transformer-based model pre-trained on paired programming and natural-language data, demonstrated strong performance when fine-tuned for statement-level summarization. Subsequent research has focused on exploiting richer structural information from source code. Several studies adopted graph-based neural networks to encode AST structures [38, 70]. Huang et al. [31] further incorporated reinforcement learning over both code tokens and AST sequences to improve statement-level summarization. With the recent success of LLMs in natural language processing, increasing attention has been directed toward leveraging them for source code summarization. Sun et al. [53] demonstrated that LLMs can generate high-quality code summaries using zero-shot prompting. Subsequent studies [1, 19] showed that few-shot prompting can significantly outperform supervised approaches. These results highlight the strong potential of LLMs to produce accurate and semantically rich code summaries. In this work, we leverage these advances to

construct a domain-adaptation dataset that provides high-quality semantic supervision for binary code comprehension.

9 Conclusion

We propose HIERONYM, a novel approach that leverages the strong generalization capabilities of LLMs to rename functions from de-compiled code in stripped binaries. To bridge the knowledge gap between general-purpose LLMs and the binary domain, we introduce a hierarchical summarization-driven domain adaptation strategy. HIERONYM further integrates multi-source information to construct comprehensive, task-relevant semantics of binary functions, thereby substantially improving function-renaming performance. In addition, we propose a dual-level evaluation framework that combines token-level and name-level metrics to enable systematic and reliable evaluation. Experimental results show that HIERONYM improves token-level precision, recall, and F1-score by 50.12%, 41.75%, and 45.10%, respectively, and achieves a 79.94% improvement in name-level accuracy over the SOTA method. Ablation studies confirm the effectiveness of key design components, and experiments on real-world malware binaries further demonstrate HIERONYM's practical utility in security analysis.

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